Zero-Shot Cross-Lingual Recognition of Sign Language Handshapes

Universitat Pompeu Fabra Barcelona

Marcel Granero-Moya* Carolina del Corral Farrarós*
Gloria Haro Coloma Ballester Ricardo Marques

Universitat Pompeu Fabra, Barcelona * equal contribution marcel.granero@upf.edu

Project page Paper

Motivation

- **Over 200 sign languages**, yet **phonological annotations exist for almost none**, Catalan Sign Language (LSC) among them.
- Handshapes can be decomposed into **five phonological features shared across languages** [1]: selected fingers, finger flexion, spread, thumb position, thumb contact.
- Goal: **zero-shot LSC handshape recognition through phonological features** predicted by models trained on American Sign Language (ASL) only, **no LSC video or label in training**.

Data and models

Dataset	Role	Handshapes	Glosses	Signers	Videos	Recording format	Aspect ratio	Frame rate
Sem-Lex [2]	ASL train & test	58	3.1k	41	85k	webcam	4:3	∼30 fps
PopSign [3]	ASL train & test	37	250	47	175k	smartphone	3:4	∼30 fps
LSC-Benchmark	LSC zero-shot test	37	–	1	74	studio	16:9	50 fps

Table 1. Datasets. ASL phonological labels are mapped through the glosses with ASL-LEX 2.0 [4]. The LSC-Benchmark is a pre-release subset with only two clips per handshape.

Three model architectures trained to predict phonological features, separately on Sem-Lex and on PopSign:

- **MLP** [5]: three fully connected layers on the 21 MediaPipe 3D hand landmarks of one least-motion frame of the dominant hand.
- **SL-GCN** [6]: spatio-temporal graph over MediaPipe pose sequences extracted from the raw videos; isolated-sign pre-training, then joint gloss and phonological features fine-tuning.
- **SHuBERT** [7]: multimodal transformer pre-trained on YouTube-ASL, fully fine-tuned for phonological feature prediction.

Zero-shot pipeline

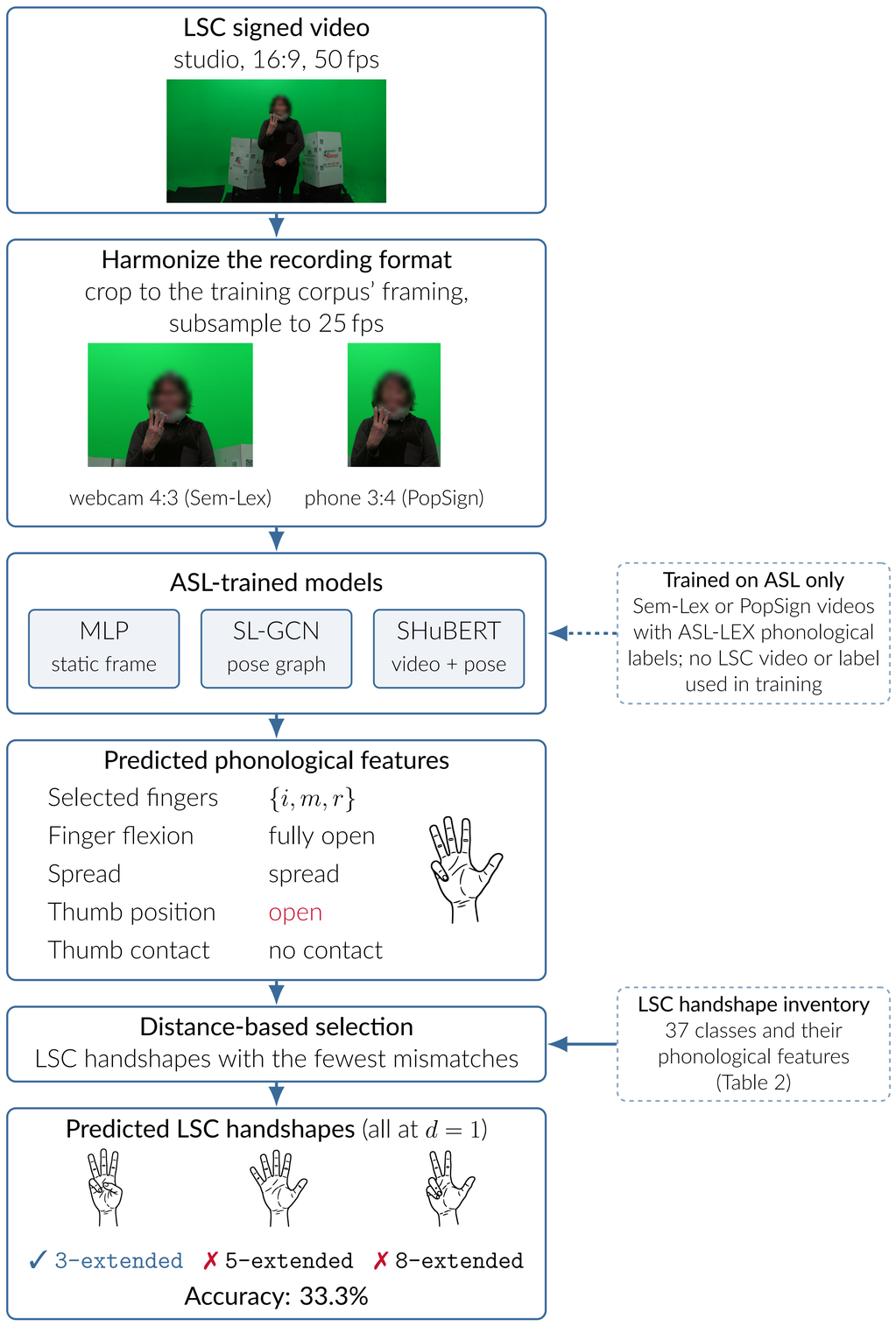


Figure 1. Zero-shot pipeline: the dashed boxes are the only language-specific inputs. Prediction example B from Table 3.

From phonological features to LSC handshapes

LSC-Benchmark handshape annotations follow the SignHub grammar [8]; we re-encode them with the five ASL-LEX phonological features:

LSC handshape	Selected fingers	Finger flexion	Spread	Thumb position	Thumb contact
`3-extended`	$\{i, m, r\}$	fully open	spread	closed	no contact
`8-extended`	$\{i, m\}$	fully open	spread	open	no contact
`5-extended`	$\{i, m, r, p\}$	fully open	spread	open	no contact
`B-extended`	$\{i, m, r, p\}$	fully open	unspread	open	no contact
`8-flat_closed`	$\{i, m\}$	flat	unspread	open	contact
⋮	*… and 32 more (37 benchmark handshapes, 43 in the grammar)*				

Table 2. LSC handshape inventory with ASL-LEX phonological features. Selected fingers: i index, m middle, r ring, p pinky.

Each phonological feature is predicted independently; the LSC handshapes with the fewest mismatched features (distance d) are the predicted handshapes:

Examples	Phonological features	Distance d
Prediction A: in the LSC inventory	$\{i, m, r\}$, fully open, spread, closed, no contact	–
→ `3-extended` (ground truth)	$\{i, m, r\}$, fully open, spread, closed, no contact	0
Predicted LSC handshapes {`3-extended`} ⇒ *accuracy*: 100%		
Prediction B: not in the LSC inventory	$\{i, m, r\}$, fully open, spread, open, no contact	–
→ `3-extended` (ground truth)	$\{i, m, r\}$, fully open, spread, closed, no contact	1
→ `5-extended`	$\{i, m, r, p\}$, fully open, spread, open, no contact	1
→ `8-extended`	$\{i, m\}$, fully open, spread, open, no contact	1
5 further handshapes at $d = 2$ (excluded)		2
Predicted LSC handshapes {`3-extended`, `5-extended`, `8-extended`} ⇒ *accuracy*: 33.3%		

Table 3. Examples of LSC handshape prediction. Blue: true handshape; red: misprediction; underlined: mismatch.

Results

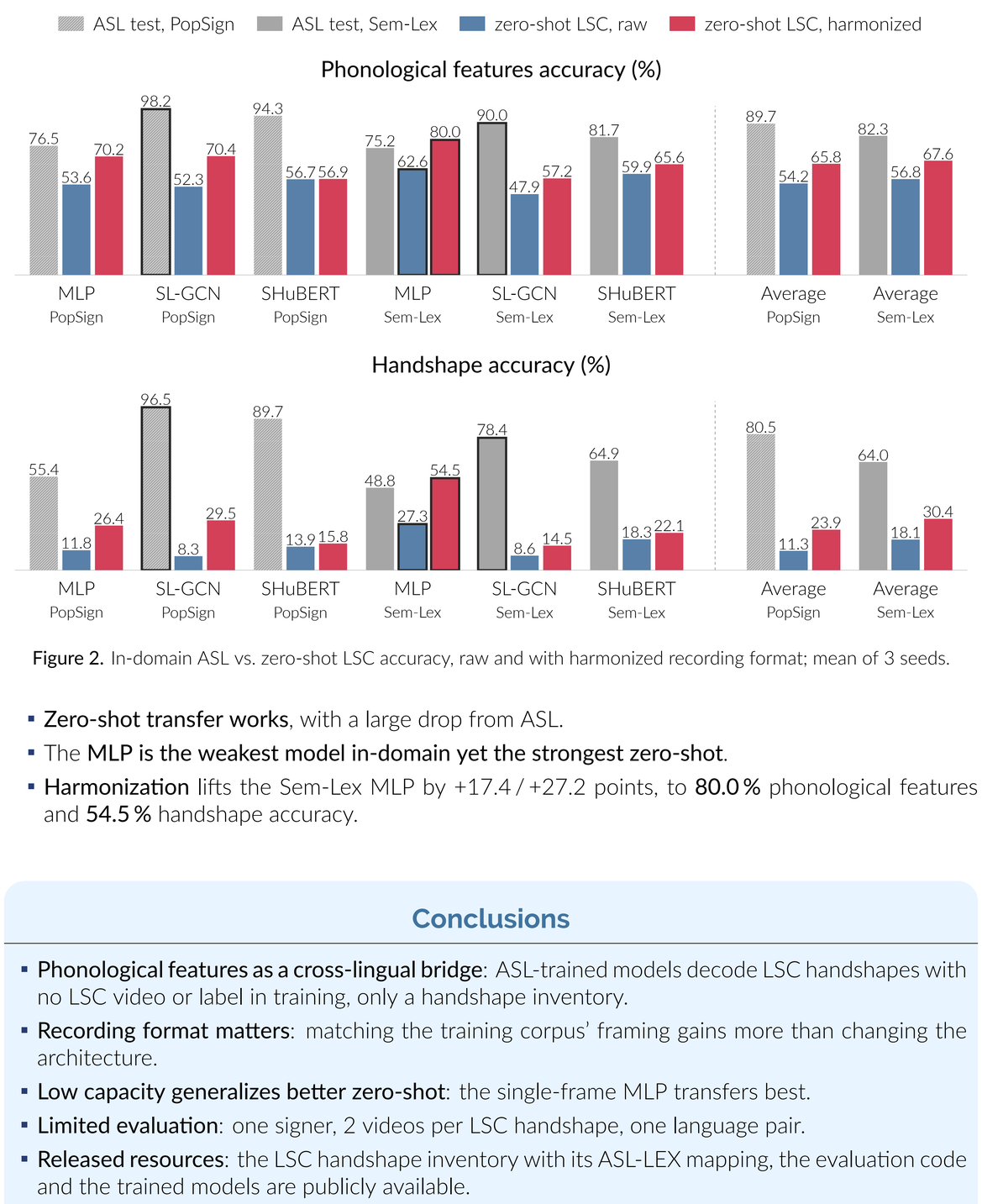


Figure 2. In-domain ASL vs. zero-shot LSC accuracy, raw and with harmonized recording format; mean of 3 seeds.

- **Zero-shot transfer works**, with a large drop from ASL.
- The **MLP is the weakest model in-domain yet the strongest zero-shot**.
- **Harmonization** lifts the Sem-Lex MLP by +17.4 / +27.2 points, to **80.0 %** phonological features and **54.5 %** handshape accuracy.

Conclusions

- **Phonological features as a cross-lingual bridge**: ASL-trained models decode LSC handshapes with no LSC video or label in training, only a handshape inventory.
- **Recording format matters**: matching the training corpus' framing gains more than changing the architecture.
- **Low capacity generalizes better zero-shot**: the single-frame MLP transfers best.
- **Limited evaluation**: one signer, 2 videos per LSC handshape, one language pair.
- **Released resources**: the LSC handshape inventory with its ASL-LEX mapping, the evaluation code and the trained models are publicly available.

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